

Article

Sustainable Management Strategies for Acarine Pests of Industrial Hemp (*Cannabis sativa* subsp. *sativa* L.)

Maria A. Canon ¹, Livia M. S. Ataide ¹, Paola Villamarin ¹, Marcello De Giosa ¹, Lance S. Osborne ², Nurhayat Tabanca ³, Sriyanka Lahiri ⁴ and Alexandra M. Revynthi ^{1,*}

¹ Tropical Research and Education Center, Institute of Food and Agricultural Sciences, University of Florida, Homestead, FL 33031, USA; malejandra1@ufl.edu (M.A.C.); lsilvaataide@ufl.edu (L.M.S.A.); wpaola.villamari@ufl.edu (P.V.); m.degiosa@ufl.edu (M.D.G.)

² Mid-Florida Research and Education Center, Institute of Food and Agricultural Sciences, University of Florida, Apopka, FL 32703, USA; lsosborn@ufl.edu

³ Subtropical Horticulture Research Station, U.S. Department of Agriculture, Agricultural Research Service (USDA-ARS), Miami, FL 33158, USA; nurhayat.tabanca@usda.gov

⁴ Gulf Coast Research and Education Center, Institute of Food and Agricultural Sciences, University of Florida, Wimauma, FL 33598, USA; lahiris@ufl.edu

* Correspondence: arevynthi@ufl.edu

Abstract

Industrial hemp (*Cannabis sativa* subsp. *sativa* L.) is an emerging crop in Florida, generating \$445 million in 2024. However, it is highly susceptible to acarine pests, including spider mites (Tetranychidae), broad mites (Tarsonemidae), and russet mites (Eriophyidae). Management options are limited due to a few federally registered products approved by the Florida Department of Agriculture and Consumer Services (FDACS-DPI). Laboratory bioassays were conducted on hemp leaf discs infested with *Tetranychus urticae*, *T. gloveri*, *Polyphagotarsonemus latus*, or *Aculops cannabicola*, and treated with biorational pesticides (citric acid, rosemary, thyme, sesame, garlic, and mineral oil) at maximum label rates. Citric acid and garlic oil were most efficacious against *T. urticae*, while garlic and thyme oils were most efficacious against the other species, causing over 80% mortality. Greenhouse trials confirmed the efficacy of citric acid and garlic oil against *T. urticae*, achieving 60–80% mortality within 24 h. Predatory mites (*Amblyseius swirskii*, *A. andersoni*, *Neoseiulus californicus*, *Galendromus occidentalis*) were evaluated against *A. cannabicola*, with *A. swirskii* showing the highest predation (≈ 20 adults/24 h) and reproduction. Compatibility tests indicated thyme and garlic oils did not significantly affect *A. swirskii* survival ($>70\%$ alive after 24 h). These findings support integrated pest management strategies for hemp acarine pests.

Keywords: botanical pesticides; integrated pest management; regulated pests; biological control; chemical control; mites



Academic Editor: Susana Pascual

Received: 28 October 2025

Revised: 21 November 2025

Accepted: 26 November 2025

Published: 2 December 2025

Citation: Canon, M.A.; Ataide, L.M.S.; Villamarin, P.; De Giosa, M.; Osborne, L.S.; Tabanca, N.; Lahiri, S.; Revynthi, A.M. Sustainable Management Strategies for Acarine Pests of Industrial Hemp (*Cannabis sativa* subsp. *sativa* L.). *Agronomy* **2025**, *15*, 2785. <https://doi.org/10.3390/agronomy15122785>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Industrial hemp, *Cannabis sativa* subsp. *sativa* L. (Rosales: Cannabaceae), hereafter referred to as hemp, is a multi-purpose plant that has been domesticated over the years to obtain fiber, seeds, medicine, essential oils, body care products, livestock food and bedding, nutritional supplements, construction materials, and textile [1]. One of the most critical characteristics of hemp, which makes it different from marijuana, is the concentration of the compound 9-tetrahydrocannabinol (THC) that must be less than 0.3% on a dry-weight basis [2]. The United States Department of Agriculture (USDA) provides the regulations for

cultivating hemp in the country [3]. In Florida this crop is regulated by the Florida Department of Agriculture and Consumer Services-Division of Plant Industry (FDACS-DPI). The hemp industry is growing in Florida in response to the growers' need to explore alternative crops that offer promising profit potential [4]. According to USDA, the value of industrial hemp production reached \$445 million, and the total planted area in the U.S. reached 45,294 acres in 2024 [5]. Given its widespread use and economic importance, several hemp cultivars have been selectively bred to enhance desired botanical characteristics such as foliage, height, and seeds [6]. These cultivars, however, are susceptible to acarine pests, and the most common in Florida include: the two-spotted spider mite, *Tetranychus urticae* Koch, the glover mite, *Tetranychus gloveri* Baker, the broad mite, *Polyphagotarsonemus latus* Banks, and the hemp russet mite, *Aculops cannabicola* Farkas [7,8]. Mites are considered major pests of hemp, causing severe infestations and damage to the plants [9]. Mite infestations can lead to leaf drop, which reduces flower and resin production, the primary products valued in the hemp industry, and can ultimately result in plant death. Moreover, mite feeding negatively affects the concentration of cannabinoids, such as THC and Cannabidiol (CBD), altering the plant's chemical composition and revealing a tradeoff between plant defense and cannabinoid levels. Consequently, mite infestations pose a serious threat to hemp production, especially when specific cannabinoid concentrations are required [10].

Tetranychus urticae is a highly polyphagous tetranychid mite well known for its rapid reproduction and ability to develop resistance to miticides [11]. *Tetranychus gloveri* is a closely related mite with similar broad feeding habits [12]. *Polyphagotarsonemus latus* is a tarsonemid mite with a worldwide distribution that frequently attacks young plant tissues [13]. More recently, the *A. cannabicola*, an eriophyid mite, has emerged as a serious pest of hemp in the USA. In Florida, *A. cannabicola* is a regulated pest by FDACS-DPI [14]. This is a minute-mite known to cause significant damage to both indoor and outdoor plants, with *C. sativa* being its only known host [15,16]. Effective management strategies for controlling mites remain largely undeveloped for hemp [16], and due to the crop's novelty, these strategies are also limited in Florida [17].

Integrated pest management (IPM) is crucial for sustainable agriculture, reducing reliance on synthetic pesticides while promoting a range of alternative pest control methods. IPM aims to optimize the effectiveness of different strategies, including cultural, mechanical, biological, and chemical control [17,18]. While chemical control is widely used for its high efficacy, availability, and cost-efficiency [19], it carries significant disadvantages, including environmental impacts, the development of pesticide resistance, and outbreaks of secondary pests [20]. Additionally, chemical control methods for hemp face specific restrictions and limitations [21], and only products approved by the state department in charge are allowed to be used on hemp. Due to the recent legalization of hemp in the state of Florida, only a limited number of registered products are available for controlling insect and mite pests on hemp [22]. Given the secondary effects of synthetic pesticides and their restrictions on hemp, there is an increasing need to adopt environmentally friendly pest management strategies. This highlights the importance of evaluating new control alternatives. Biorational pesticides present a promising solution, offering advantages such as specificity, less detrimental effects to non-target species, low toxicity to humans and the environment, rapid decomposition, and compatibility with other management practices, these characteristics make biorational pesticides an ideal option for pest control [23]. Botanical insecticides are a category of biorational pesticides that derive from plant-based sources [24]. These insecticides have gained worldwide attention, especially in organic farming, due to their numerous benefits and their role in sustainable IPM programs targeting soft-bodied pests of fruits, shade trees, and ornamental plants [24,25]. These products exhibit various modes of action, including antifeedant, molting, and respiratory inhibition, and repellent effects, which

have contributed to a significant reduction in synthetic pesticide use and encouraged research interest in recent years [26]. However, further studies are needed to strengthen user confidence.

Biological control is another important pest management strategy to explore and integrate into IPM programs developed for hemp production. This management strategy's benefits are its target specificity, minimal environmental impact, and low toxicity [27]. Predatory mites, from the family Phytoseiidae (Acari: Mesostigmata), are the most used biocontrol agents against phytophagous mites, mainly in vegetable and ornamental production [28]. Phytoseiids are key natural enemies of major pests, including thrips, whiteflies, and phytophagous mites, and are widely employed in both outdoor and greenhouse crops [29].

Chemical control, using botanical pesticides, combined with biological control, represents a promising biorational strategy for managing acarine pests on hemp. However, the potential compatibility between these two approaches remains a concern, as several factors can make natural enemies susceptible to botanical pesticides. These include the developmental stage of the natural enemy, application rate, and timing of application, as well as the species and types of biological control agent (parasitoid or predator) [30,31]. This highlights the importance of evaluating the effects of selected biorationals on the intended biological control agents. Integrating both strategies is essential for successfully managing mite pests.

Considering the novelty of hemp as a crop, and the significant knowledge gaps that still exist in acarine pest management, particularly regarding effective control tools and integrated strategies, this study aimed to identify efficacious botanical pesticides against the main acarine pests of hemp and assess the potential of biological control agents for sustainable mite management. Single-active-ingredient botanical pesticides (citric acid, garlic, sesame, thyme, rosemary oils) with Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) 25(b) United States Environmental Protection Agency (EPA) exemption were selected from the FDACS-DPI approved list to be tested against the most common acarine pests in Florida. All botanical products evaluated are expected to be effective against all developmental stages of the acarine species, because they are labeled as miticides.

Furthermore, this study explored the compatibility of chemical and biological control methods to ensure their integration without compromising each other's efficacy. To this end, commercially available predatory mites, including *Amblyseius swirskii* (Athias-Henriot), *Amblyseius andersoni* (Chant), *Neoseiulus californicus* (McGregor), and *Galendromus occidentalis* (Nesbitt) were evaluated for their potential biological control against *A. cannabicola*. All predatory mites are expected to exhibit high predation and oviposition rates when feeding on *A. cannabicola*, since they all have been previously reported as predators of eriophyid mites [32,33].

Together, these results provide baseline information on chemical and biological options and offer a foundation for developing future integrated and sustainable management strategies for acarine pests in Florida's emerging hemp industry.

2. Materials and Methods

2.1. Host Plants

The host plants were grown from commercially sourced seeds, with the cultivar/variety information provided on the seed label. Pepper plants, *Capsicum annuum* L. (Solanales: Solanaceae), 'Rainbow blend' (Seed Needs LLC, Michigan, IN, USA) and bean plants, *Phaseolus vulgaris* L. variety 'Roman' (Goya Foods, Inc.[®], New York, NY, USA) were grown from seeds in plastic 140 mL pots filled with ProMix soil (ProMix BX Mycorrhizae, Denver, CO, USA). The pots were placed in a room with controlled conditions

(25 ± 1 °C, $50 \pm 10\%$ RH, and 12:12 h L:D photoperiod) for seed germination. The plants were watered three times a week, and no fertilizer was added. Five-week-old pepper plants and two-week-old bean plants were used to maintain mite colonies. Hemp plants, *C. sativa* variety G 33211 21UO SD ‘wife’, were grown from cuttings. In a growth chamber (Panasonic Versatile Environmental Test Chamber MLR-352H), hemp plants were watered daily and fertilized once a week with a 24N-8P-16K solution (Miracle-Gro, Marysville, OH, USA) and maintained under controlled conditions (27 ± 1 °C, $70 \pm 10\%$ RH, and 16:8 h L:D photoperiod).

2.2. Pest Mite Colonies

Mites were identified using both the available original descriptions and the redescrptions for each species, and the identifications were confirmed by FDACS-DPI. *Aculops cannabicola*: The mites were obtained from an established colony under the permit number 2021-027 provided by FDACS-DPI and maintained on hemp (*C. sativa*) plants from the ‘wife’ cultivar for multiple generations. The colony was kept inside a mite-proof mesh cage (W 32.5 cm $\times$ D 32.5 cm $\times$ H 77.0 cm, mesh diameter 160 μ m; BD4E3074 Bugdorm, Taichung, Taiwan) inside a growth chamber at 27 ± 1 °C, 60% RH, and 16:8 h L:D photoperiod and provided with clean and fresh plants monthly. The plants were watered four times per week.

Polyphagotarsonemus latus: The mites were obtained from an established colony and maintained for multiple generations on bell pepper plants (*C. annuum*) grown from seeds. The colony was supplemented with fresh and clean plants when the symptoms and damage on infested plants were obvious (approx. once a month). The plants were watered three times a week, and no fertilizer was added. The colony was maintained in a mite-proof mesh inside a growth chamber at 27 ± 1 °C, 70% RH, and 12:12 h L:D photoperiod.

Tetranychus urticae and *T. gloveri*: The mites were obtained from an established colony and maintained for multiple generations on bean plants [11]. The colonies were supplemented with clean plants three times per week. Each colony was maintained in a mite-proof mesh cage inside a growth chamber at 27 ± 1 °C, 50% RH, and 12:12 h L:D photoperiod.

Predatory mites: The predatory mite species *N. californicus*, *A. swirskii*, *A. andersoni*, and *G. occidentalis* were sourced from commercial suppliers (Biobee, Salisbury, MD, USA, Koppert, Howell, MI, USA and Beneficial Insectary, Redding, CA, USA). Upon arrival, each species was maintained separately in custom made, mite-proof plastic containers (9.5 cm high $\times$ 27 cm long $\times$ 19 cm wide). Within each container, a walking arena was constructed using a black plastic sheet (17 cm $\times$ 9 cm) placed on top of a water-saturated foam base (3.5 cm high $\times$ 17.5 cm long $\times$ 9 cm wide) (3M, Saint Paul, MN, USA) and surrounded by water-saturated cotton tissue (Fisherbrand, Pittsburgh, PA, USA), following the standard phytoseiid mite rearing protocol described by Overmeer [34]. Two bean leaves infested with *T. urticae* obtained from the colony and *Typha* spp. pollen (Nutrimite, Biobest, Victor, CA, USA) were provided as a food source twice per week. Colonies were maintained in an incubator at 27 ± 1 °C, $80 \pm 5\%$ RH, and 12:12 h L:D photoperiod.

2.3. Botanical Pesticides—Laboratory Bioassays

To screen registered botanical pesticides approved for hemp use, a laboratory bioassay was conducted. The experimental arena consisted of a Petri dish (5 cm diameter) (Fisherbrand, Pittsburgh, PA, USA) with a layer of agar 1% (Difco™ Agar Bacteriological, Winsor & Newton, London, UK). A hemp leaf disc (1 cm diameter) from a clean plant was cut using a cork borer and positioned with the abaxial surface facing downward in the middle of the Petri dish. This medium was used to prevent mites from drowning. Ten Petri dishes were prepared for each mite species (*A. cannabicola*, *T. urticae*, *T. gloveri*, and

P. latus), and ten adult females of unknown age were collected from the stock colonies and transferred on each leaf disc using a fine brush (CotmanTM, Winsor & Newton, London, UK). Each Petri dish with ten mites represented one replicate. The botanical pesticides (garlic oil, thyme oil, rosemary oil, sesame oil, mineral oil, and citric acid) were mixed with water at the maximum label rate (Table 1). Solutions of 500 mL were prepared for each pesticide. Each pesticide solution or water control (30 mL) were placed in plastic sprayer containers (Central Pneumatic, Calabasa, CA, USA) and connected through a hose to the compressor at 30 PSI (Porter Cable, Jackson, TN, USA). Subsequently, the leaf discs with the mites were directly sprayed with the solution, maintaining 20 cm of distance until droplets were evident on the leaf surface, approx 1–2 s. The sprayed experimental arenas were allowed to dry for 1 h, after which they were placed inside a growth chamber at 25 °C, RH 60%, 12:12 h L:D photoperiod. Evaluations were conducted 4, 24, 48, 72, and 96 h after the treatment application, where mortality and oviposition were scored. Mortality was assessed when exposed mites showed no movement of any body appendage paired with a shriveled body observed under the stereomicroscope (Leica Stereozoom S9E, Heerbrugg, Switzerland). In total, ten replicates per mite species and treatment were conducted. All mites and treatments were tested simultaneously once. The experiment followed a completely randomized design.

Table 1. Treatments tested against *Tetranychus urticae*, *Tetranychus gloveri*, *Polyphagotarsonemus latus* and *Aculops cannabicola*.

Product	Active Ingredient	Rate/ha	Solution	EPA Registration Number
Bush Doctor Force of Nature Insect Repellent	Garlic oil	1.53 L/ha	11.8 mL/L	FIFRA 25b-exempt
Nuke EM	Citric Acid	0.1 L/ha	62.5 mL/L	FIFRA 25b-exempt
Organocide bee safe 3-in-1 garden spray concentrate	Sesame oil	0.04 L/ha	23.4 mL/L	FIFRA 25b-exempt
SNS 217C	Rosemary oil	0.65 L/ha	5 mL/L	FIFRA 25b-exempt
Thyme Guard	Thyme oil	5%	50 mL/L	FIFRA 25b-exempt
Suffoil-X	Mineral oil	2%	20 mL/L	48813-1-68539
	Water	NA	NA	NA

NA: Not applicable.

2.4. Botanical Pesticides—Greenhouse Experiments

Following the laboratory bioassays, greenhouse experiments were conducted to assess the most efficacious products against *T. urticae*. For these evaluations *T. urticae* was used as it is the most common and persistent pest mite species in South Florida [14]. Pest-free three-month-old hemp plants (30 cm tall) were individually placed inside mite-proof mesh cages (W 32.5 cm × D 32.5 cm × H 77.0 cm, mesh diameter 160 µm; BD4E3074 Bugdorm, Taichung, Taiwan), and each plant represented one replicate. Thirty (two-day-old) gravid females were released onto each isolated plant, and the mites were allowed to establish for six weeks. After this time, populations were estimated on the plants by scoring all developmental stages: eggs, larvae, nymphs, and adults. Five leaves from each plant were randomly selected and examined under a stereomicroscope. Subsequently, garlic oil, citric acid at the maximum label rate (Table 1), and water (control) were applied to each plant using a pump sprayer (HDX, Mansfield, OH, USA) until runoff. Evaluations were conducted before treatment application (prior), 24 h, and seven days after the application by randomly collecting five leaves from each plant and examining them under a stereomicroscope. The mite population on each leaf was recorded, including all developmental stages: eggs,

larvae, nymphs, and adults. The experiment was performed under controlled conditions (27 ± 2 °C, $70 \pm 15\%$ RH, and a 16:8 h L:D), consisted of ten replicates per treatment, and it was repeated twice (in blocks). The experiment followed a completely randomized design.

2.5. Biological Control of *A. cannabicola*—Laboratory Bioassays

Before initiating the predation assays, the ability of *N. californicus*, *A. swirskii*, *A. andersoni*, and *G. occidentalis* to prey on *A. cannabicola* was confirmed. These predator species were selected because they are able to feed and reproduce on eriophyoids [32,33]. A no-choice experiment was then conducted in which each female predator mite was offered either 30 adult *A. cannabicola* (treatment) or 60 *T. urticae* eggs (control) on a hemp leaf disc (diameter = 0.3 mm). Each disc was placed on 1% agar inside a transparent 3 mL acrylic container fitted with screw lids modified to include a mite-proof mesh (54 µm) for ventilation. Prior to the experiment, predatory mites were starved for 24 h. Prey and predators were manually transferred to the leaf discs using a fine brush. Prey consumption by *A. swirskii* and *A. andersoni* was assessed after 24 h. Due to the low predation rates observed during the first 24 h for *N. californicus* and *G. occidentalis*, prey consumption was evaluated after 48 h for these two predators. All predator mite species were tested using the same set-up ($n = 20$ replicates per predator mite species). The experiment followed a completely randomized design. All arenas were maintained in a growth chamber at 27 °C, 80% RH, and a 12:12 h L:D photoperiod.

2.6. Compatibility Bioassays

The most efficacious botanical pesticides against *A. cannabicola* were evaluated to determine their effect on *A. swirskii*. Each product was tested at its maximum labeled rate (Table 1). For each treatment, 500 mL of solution was prepared, and 20 mL aliquots were transferred to handheld sprayers (Central Pneumatic, Calabasas, CA, USA) connected to a compressor set at 30 PSI. A water-only treatment served as the control. The bioassay arena was the same as described above: one *A. swirskii* female of unknown age was transferred from the colony to a bean leaf disc ($d = 0.3$ mm) placed on a 1% agar substrate inside the acrylic container. Each leaf disc containing one female predator was directly sprayed with approximately 0.2 mL of the test solution as described in Section 2.3. Immediately after spraying, the containers were sealed to prevent mites from escaping. Predator mortality was assessed at 24-, 48-, 72-, and 96 h post-treatment ($n = 35$ replicates per treatment). Fecundity data were not collected. The experiments were repeated twice in different times (blocks). The experiment followed a completely randomized design. All arenas were maintained in a growth chamber at 27 °C, 80% RH, and a 12:12 h photoperiod.

2.7. Statistical Analysis

Data analysis was performed in R version 4.2.2. Model selection was based on the Akaike Information Criterion (AIC), residual diagnostics based on R plots, and simulation-based diagnostics from the DHARMA package [35]. For the laboratory experiments targeting acarine pests, the mortality of the mites was assessed with a time-to-event analysis with a parametric model with a Gaussian error distribution. The factors included seven treatments (citric acid, rosemary, thyme, sesame, garlic, mineral oil products, and water control). Post hoc tests were performed using the estimated marginal means (EMMs) with Tukey's HSD adjustment for the probabilities, implemented with the emmeans package [36]. For the greenhouse experiments, the proportion of dead mites (mortality) was analyzed using a generalized linear mixed-effects model (GLMM) with a binomial error distribution implemented with the glmer function of the lme4 package [37]. The model included the following fixed effects: 'treatments' (citric acid, garlic oil products, and water control), 'time' (prior, 24 h, 7 days), and their 'interactions.' Random effects included the

blocks. Post hoc tests were performed using EMMs. For the oviposition, the number of eggs was fitted in a GLMM with a negative binomial error distribution. The model included the same fixed and random effects as the model used for the mortality data. Post hoc tests were also performed similarly to the mortality data.

Regarding the biological control assays, the number of prey consumed per predatory mite species was analyzed using generalized linear model (GLM) with a quasipoisson error distribution. Each predatory mite species was analyzed separately and included in the model as categorical variable while the number of prey consumed was included as response variable. The survival of *A. swirskii* was assessed with a time-to-event analysis with a parametric model and a Gaussian error distribution. The factors included five treatments (citric acid, thyme oil, garlic oil, mineral oil, and water control). Post hoc tests were performed using EMMs.

3. Results

3.1. Botanical Pesticides—Laboratory Bioassays

Direct spray of all botanical pesticides on adult females caused significant mortality to *T. urticae* (parametric model: $\chi^2 = 259.68$; d.f. = 6; $p < 0.001$, Figure 1A), *T. gloveri* (parametric model: $\chi^2 = 359.1$; d.f. = 6; $p < 0.001$, Figure 1B), *P. latus* (parametric model: $\chi^2 = 434.53$; d.f. = 6; $p < 0.001$, Figure 1C) and *A. cannabicola* (parametric model: $\chi^2 = 31.52$; d.f. = 6; $p < 0.001$, Figure 1D). After 96 h, garlic oil was the most efficacious treatment, causing the highest mortality to all the species, with mortality exceeding 80%. The following treatment that effectively caused mortality was thyme oil in all the species exposed to this product after 96 h, except for *T. urticae*, where citric acid, followed by garlic oil, caused the highest mortality of around 80% ($\pm 0.028\%$). Mortality in the water was lower than 20% ($\pm 0.033\%$) for all the mites except for *T. urticae* that reached 58% ($\pm 0.048\%$) after 96 h (Figure 1).

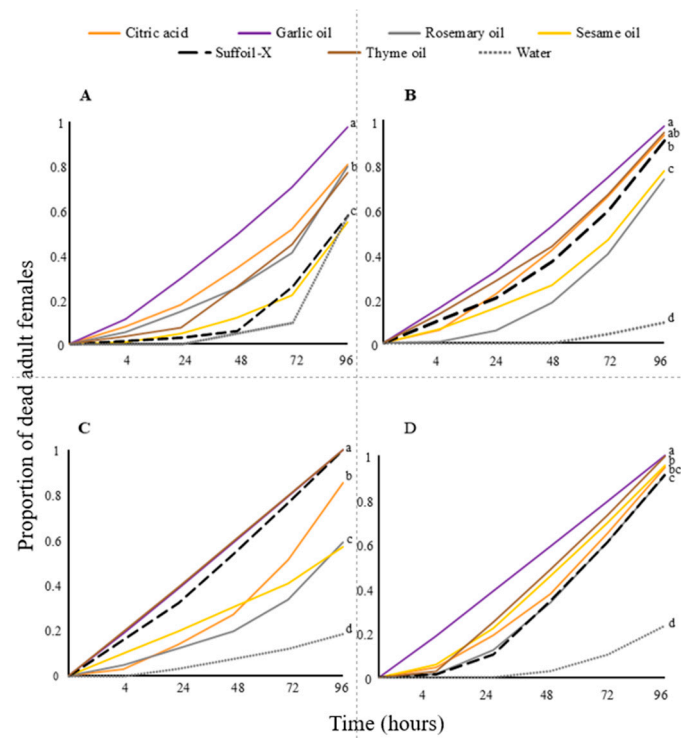


Figure 1. Proportion of dead adult females caused by botanical pesticides over the experimental time (4, 24, 48, 72, and 96 h). (A) *Tetranychus urticae*, (B) *Tetranychus gloveri*, (C) *Polyphagotarsonemus latus*, and (D) *Aculops cannabicola*. The lines with different lowercase letters indicate significant differences among treatments (Parametric model, $p < 0.05$, $n = 10$ per treatment).

3.2. Botanical Pesticides—Greenhouse Experiments

Mortality was significantly affected by the interaction of treatments and time (Table 2; Figure 2A). Prior to treatment application, mite populations were uniform among plants, and the proportion of mite mortality was very low across treatments and the control. Pairwise comparisons showed that citric acid (0.009 ± 0.022) (Mean proportion of dead mites, $\pm$ SE) resulted in a significantly higher proportion of mite mortality compared to garlic oil (0.003 ± 0.012 ; $p < 0.0001$) and water (0.002 ± 0.010 ; $p < 0.0001$). After 24 h garlic oil (0.811 ± 0.087) application resulted in significantly higher mortality than citric acid (0.621 ± 0.108 ; $p < 0.0001$) and water (0.005 ± 0.016 ; $p < 0.0001$). Similarly, citric acid application caused significantly higher mortality than water ($p < 0.0001$). Seven days after the treatment application, garlic oil (0.404 ± 0.109) caused significantly higher mortality than citric acid (0.335 ± 0.105 ; $p < 0.0001$) and water (0.004 ± 0.014 ; $p < 0.0001$), while citric acid caused significantly higher mortality than water ($p < 0.0001$) (Figure 2A).

Table 2. Treatment, time and interaction effects on the *Tetranychus urticae* mortality and oviposition. Data were analyzed using a Generalized linear mixed-effects model.

Response Variable	Treatment Effect	Time Effect	Interaction Effect
Mite mortality	$\chi^2 = 3466.19$; d.f. = 2; $p < 0.001$	$\chi^2 = 3466.84$; d.f. = 2; $p < 0.001$	$\chi^2 = 201.93$; d.f. = 4; $p < 0.001$
Mite oviposition	$\chi^2 = 81.64$; d.f. = 2; $p < 0.001$	$\chi^2 = 329.16$; d.f. = 2; $p < 0.001$	$\chi^2 = 156.08$; d.f. = 4; $p < 0.001$

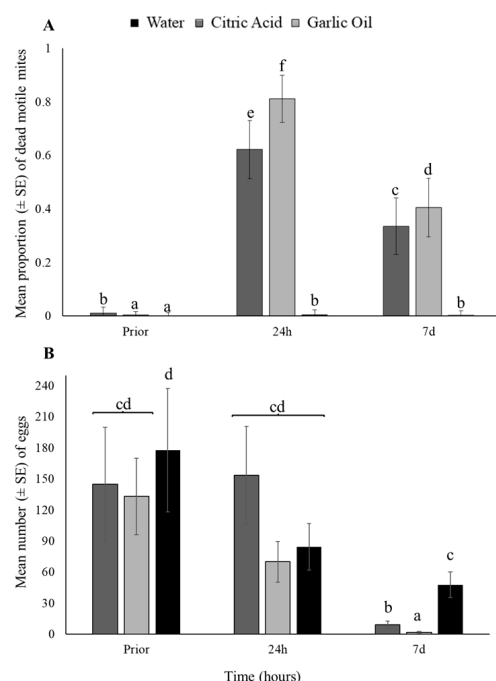


Figure 2. (A) Mean proportion ($\pm$ SE) of dead *Tetranychus urticae* motiles caused by botanical pesticides (citric acid and garlic oil) over the experimental time (prior to the treatment exposure, 24 h, and 7 days after the treatment exposure). (B) Mean number of *T. urticae* eggs ($\pm$ SE) over the experimental time prior. Lowercase letters indicate significant differences among all treatments due to significant interaction between treatment and time (EMMs, $p < 0.05$, $n = 20$).

Oviposition was also significantly affected by the interaction of treatments and time (Table 2; Figure 2B). Prior to treatment exposure, there were no significant differences in the number of eggs among the treatment groups and the control (garlic oil: 133 ± 36.9 ; citric acid: 144.7 ± 54.9 ; water: 177.7 ± 59.5). Similarly, after 24 h of exposure to citric acid (153.6 ± 47.1) did not differ significantly from garlic oil (70.0 ± 19.7 ; $p = 0.05$) or water (84.4 ± 22.6 ; $p = 0.1$), and garlic oil and water also showed no significant difference

($p = 0.95$). Seven days after mite exposure to citric acid (9.15 ± 3.58) resulted in significantly more eggs than exposure to garlic oil (1.52 ± 0.92 ; $p < 0.0001$) but significantly fewer eggs than water (47.7 ± 12.42 ; $p < 0.0001$). Exposure to garlic oil also resulted in significantly fewer eggs than water ($p < 0.0001$) (Figure 2B).

3.3. Biological Control of *A. cannabicola*—Laboratory Bioassays

All predatory mite species tested showed a higher predation on *T. urticae* compared to *A. cannabicola*. *Amblyseius swirskii* consumed significantly more *T. urticae* eggs (30 ± 2 ; mean $\pm$ SE) than *A. cannabicola* adults (20 ± 1) within 24 h (GLM: $F = 30.5$; d.f. = 1; $p < 0.001$; Figure 3A), while predator oviposition among treatments was not statistically different (GLM: $F = 2.05$; d.f. = 1; $p = 0.16$; Figure 4A). Similarly, *A. andersoni* preyed significantly more on *T. urticae* (27 ± 3) than on *A. cannabicola* (13 ± 2) within 24 h (GLM: $F = 19.0$; d.f. = 1; $p < 0.001$; Figure 3B); however, no predator eggs were observed in any of the treatments. After 48 h, *N. californicus* also consumed significantly more *T. urticae* eggs (26 ± 2) than *A. cannabicola* adults (13 ± 2) (GLM: $F = 13.4$; d.f. = 1; $p = 0.001$; Figure 3C). Likewise, *G. occidentalis* fed on significantly more *T. urticae* eggs (35 ± 3) than *A. cannabicola* adults (17 ± 2) over 48 h (GLM: $F = 32.5$; d.f. = 1; $p < 0.001$; Figure 3D).

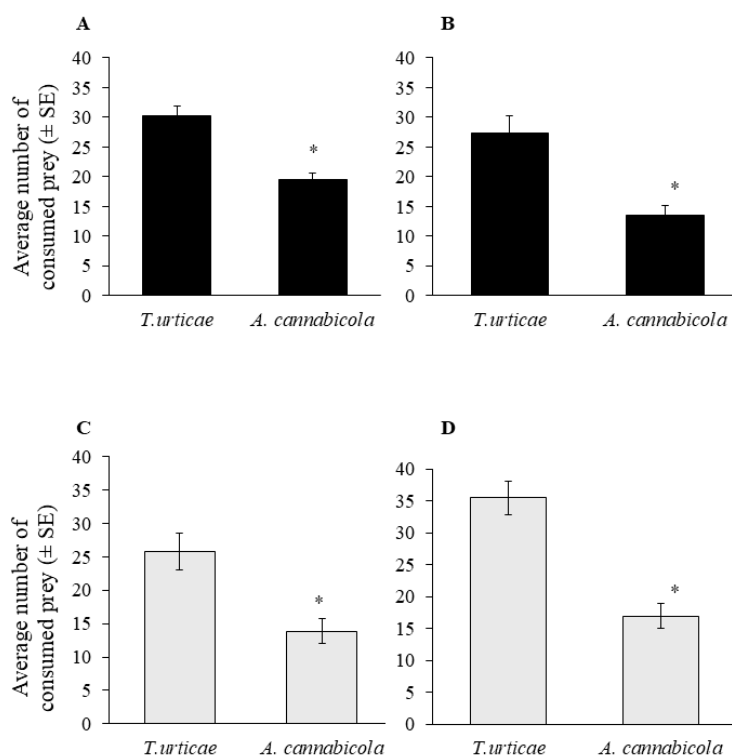


Figure 3. Average number of consumed prey ($\pm$ SE) by four species of predatory mites when offered either *Tetranychus urticae* eggs or *Aculops cannabicola* adults in a no-choice assay. Predation of (A) *Amblyseius swirskii* and (B) *Amblyseius andersoni* was assessed after 24 h. Predation of (C) *Neoseiulus californicus* and (D) *Galendromus occidentalis* was assessed after 48 h due to initially low predation. Asterisks represent significant differences among the treatments (EMMs, $p < 0.05$, $n = 20$).

After 48 h, *N. californicus* also consumed significantly more *T. urticae* eggs (26 ± 2) than *A. cannabicola* adults (13 ± 2) (GLM: $F = 13.4$; d.f. = 1; $p = 0.001$; Figure 3C). During this period, predator oviposition was only observed when feeding on *T. urticae* eggs (GLM: $F = 38.0$; d.f. = 1; $p < 0.001$; Figure 4B). *Galendromus occidentalis* fed on significantly more *T. urticae* eggs (35 ± 3) than *A. cannabicola* adults (17 ± 2) over 48 h (GLM: $F = 32.5$; d.f. = 1; $p < 0.001$; Figure 3D). In addition, predators laid fewer eggs after feeding on *A. cannabicola* compared to *T. urticae* (GLM: $F = 14.0$; d.f. = 1; $p < 0.001$; Figure 4C).

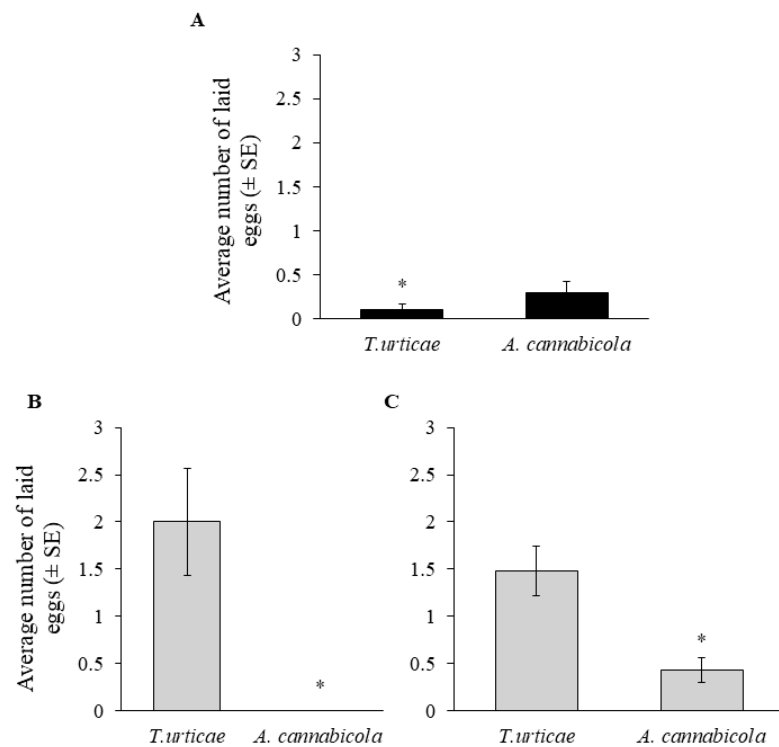


Figure 4. Average number of laid eggs ($\pm$ SE) by the three species of predatory mites when offered either *Tetranychus urticae* eggs or *Aculops cannabicola* adults in a non-choice assay. Oviposition of (A) *Amblyseius swirskii* was assessed after 24 h while oviposition for (B) *Neoseiulus californicus* and (C) *Galendromus occidentalis* was assessed after 48 h. Asterisks represent significant differences among the treatments (EMMs, $p < 0.05$, $n = 20$).

3.4. Compatibility Bioassays

Direct spray of biorational pesticides on *A. swirskii* significantly affected its survival over time (parametric model: deviance = 17.1; d.f. = 4; $p = 0.002$; Figure 5). Citric acid and mineral oil significantly reduced the survival of *A. swirskii* compared to thyme oil, garlic oil, and the water control.

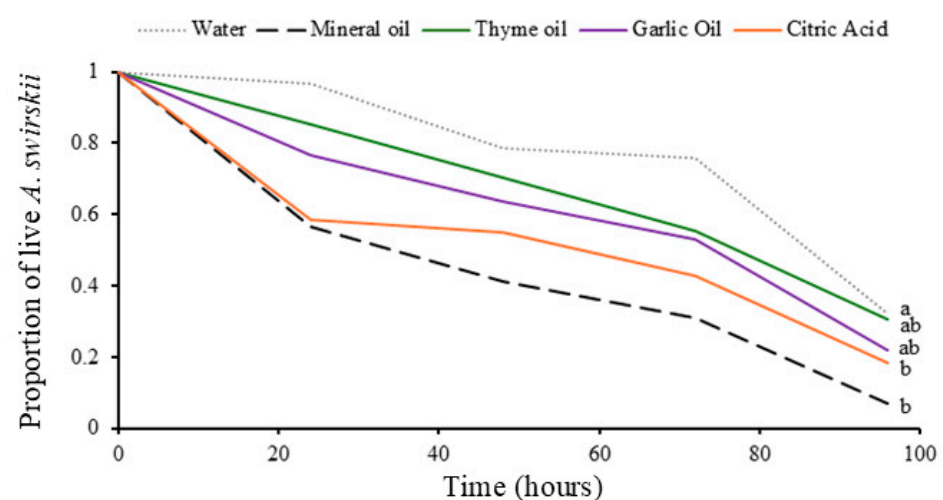


Figure 5. Average proportion of live *Amblyseius swirskii* after direct exposure to botanical pesticides, citric acid, mineral oil, thyme oil, garlic oil, and water control over 96 h. Each treatment was applied at its maximum label rate. No supplemental food was provided during the experiment. Different letters represent significant differences among the treatments (parametric model, $p < 0.05$, $n = 35$).

4. Discussion

In the present study, the first aim was to identify efficacious botanical pesticides against major acarine pests of hemp. Our results identified three promising products: garlic oil, citric acid, and thyme oil as the most efficacious against adult female of *T. urticae*, *T. gloveri*, *P. latus*, and *A. cannabicola*. Laboratory assays demonstrated that garlic oil exhibited the highest efficacy, resulting in nearly 100% mortality, while thyme oil and citric acid caused more than 80% of the mite mortality by the end of the experiment (Figure 1). Subsequent greenhouse evaluations of garlic oil and citric acid on hemp plants against *T. urticae*, corroborated the laboratory bioassays results, with both treatments causing significant mite mortality (Figure 2A) and reduced oviposition (Figure 2B). Overall, the findings of this study fill the existing knowledge gaps regarding the management of acarine pests of hemp and pave the way for the development of a more sustainable IPM program.

Based on the laboratory findings, the most efficacious products against *T. urticae* were further evaluated under more realistic conditions on hemp plants in a greenhouse setting. *Tetranychus urticae*, is the most pressing pest in indoor [38] and outdoor [14] hemp production, and therefore, it was selected for greenhouse evaluations. The results demonstrated that garlic oil caused the highest mortality, eliminating over 80% of the motile population within 24 h of treatment application, followed by citric acid with over 60% mortality of the motile population (Figure 2A). Regarding *T. urticae* oviposition, the number of eggs declined over the course of the experiment in both treatments and the control. In the treatments, this egg decrease was likely due to a reduction in the adult female population caused by the applied products, rather than an ovicidal effect of the pesticides. In contrast, the decline in egg production observed in the control may be attributed to the overexploitation of the food source, which likely triggered dispersal behavior. When the environment becomes hostile or non-viable, individuals must disperse in search of better conditions. This dispersal behavior is driven largely by the availability of key resources such as food, which plays a crucial role in mite survival [39,40]. In the current greenhouse experiments, we observed adult *T. urticae* females congregating at the corners of the cages, suggesting an attempt to disperse in search of new food sources.

Horticultural oils and soaps can control arthropod pests by obstructing the spiracles, inhibiting gaseous exchange, and causing death due to asphyxiation. Additionally, they may cause metabolic disorders by intoxication [24]. The effects caused by these products are closely linked to their chemical composition; for instance, horticultural oils have a narrow distillation range, which facilitates their application on leaves and reduces the risk of phytotoxicity [25]. These plant-derived oils contain fatty acids and other lipids, along with refined petroleum components that are usually combined with emulsifying agents to enable mixing with water [41,42]. Soaps, on the other hand, are composed of potassium salts of a plant-derived fatty acid with long carbon chains [25]. Many studies have evaluated the acaricidal activity of garlic extracts against phytophagous mites, and this activity could be due to the presence of dipropyl disulfide and dipropyl trisulfide [43]. Garlic oil was reported to be efficacious in controlling the red spider mite, *Oligonychus coffeae* Nietner [44], *Acalitus simplex* Flechtmann et Etienne [45], and the two-spotted spider mite, *T. urticae*, impacting their fecundity and fertility [26]. The toxic effects of garlic oil are related to the high concentrations of organosulfur compounds, which have been proven to reduce mite pest populations [46]. The most abundant components of garlic extracts are diallyl trisulphide and diallyl disulphide, which have a highly reactive nature and may cause inhibition by cross-linking with essential thiol compounds in enzyme structures, altering the protein and denaturing it [47]. Citrus essential oils have been shown to effectively manage *T. urticae* by influencing the pest's behavior, particularly in terms of feeding and egg-laying preferences. The most abundant components of citrus oils are monoterpenes, and they may be related to

the negative effects caused on this pest [48]. Building on these findings, botanical pesticides offer an attractive alternative to conventional chemical pesticides. Unlike conventional pesticides, they provide several benefits, such as rapid degradation in the environment, minimal bioaccumulation, and reduced toxicity to humans [20]. This is particularly important given the persistent development of resistance to conventional pesticides by *T. urticae*. For example, Döker et al. [11] demonstrated that *T. urticae* populations in South Florida have developed resistance to three commonly used active ingredients, including abamectin, pyridaben, and cyflumetofen. Collectively, these observations highlight botanical pesticides as a promising avenue for *T. urticae* management and control. Despite the advantages of botanical pesticides, phytotoxicity is often a concern. However, no phytotoxic effects were observed in the treated plants. The chlorosis and yellowing symptoms present were attributed to mite feeding and appeared in both the treated and control plants.

IPM employs multiple strategies apart from chemical control. One such strategy is biological control, which has been extensively studied for *T. urticae* due to its worldwide distribution, broad host range, and significant economic impact on various agricultural crops [49]. The release of *Phytoseiulus persimilis* Athias-Henriot to control *T. urticae* on hemp has already been demonstrated. In contrast, research on *A. cannabicola* is limited, largely due to the legal restrictions surrounding *C. sativa* [50]. This study evaluated four commercially available predators that have the potential to be used in the biological control of eriophyids, because they are classified as type II or III based on McMurty et al. [32]. Although these predatory mites are not specialists of eriophyids, some have been evaluated against various eriophyid species. For example, *A. andersoni* is classified as type III-b lifestyle, which means that it may feed and reproduce on a wide range of prey and inhabit glabrous leaves of deciduous plants. *Galendromus occidentalis* is classified as type II, which means this mite primarily feeds on *Tetranychus* species, but it may feed in other mite groups, and both predatory mites can prey on *Aculus schlechtendali* Nalepa. *Neoseiulus californicus* is also classified as type II and can prey on *Aculops lycopersici* Tryon [33,51,52]. The two predatory mite species, *N. californicus* and *G. occidentalis*, exhibited relatively low predation rates compared to the other evaluated species, consuming fewer than 20 individuals after 48 h. This may be explained by their classification as type II predators [33]. Classified as a Type III generalist predator, *A. swirskii* may feed and reproduce on a wide range of prey [28]. This predator has previously been identified as a potential biological control agent against another important eriophyid species, *A. lycopersici*, as it is capable of feeding, reproducing, and establishing populations when this mite is available [53,54]. Our results showed that *A. swirskii* can feed and reproduce on *A. cannabicola*. Both *A. cannabicola* and *A. lycopersici* share a similar ecological lifestyle, they are vagrant mites whose entire life cycle occurs exposed on the leaf surface and plant structures, making them more vulnerable to predatory mites, unlike other eriophyid species that form galls or seek refuge [33]. The ability of *A. swirskii* to survive and reproduce on alternative food sources, such as pollen [55], suggests a potential use as a preventive control strategy by establishing predator populations on hemp prior to pest infestation. Although *A. swirskii* has shown promising results under laboratory conditions, this does not necessarily ensure effective control of the pest in greenhouse or field environments. In real-world scenarios, the pest is distributed throughout the entire plant, and plant features influence biological control outcomes [56]. Hemp plants possess both simple and glandular trichomes that secrete resin [57], which may affect the effectiveness of the predatory mites, as happened with *A. swirskii* against *A. lycopersici* on tomato plants, where the glandular trichomes of the plant affected the success of the predatory mite [58]. Greenhouse experiments were not permitted by FDACS-DPI and therefore, further research under more realistic conditions is needed to assess the true potential of this predatory mite against *A. cannabicola*.

Another predatory mite species that showed potential for feeding on *A. cannabicola* was *A. andersoni* [59]. However, unlike *A. swirskii*, *N. californicus*, and *G. occidentalis*, which were able to lay eggs when provided with *A. cannabicola* as a food source, *A. andersoni* did not oviposit while feeding on this prey. Oviposition is a crucial factor when choosing a biocontrol agent, since its ability to reproduce and establish contributes to population growth over time [60]. The integration of chemical and biological control strategies has been a long-term concern, and, over the past decade, their compatibility has been increasingly studied. The toxicity of a pesticide to predatory mites (phytoseiids) is determined by critical physiological processes, including toxicokinetics and toxicodynamics—meaning how the chemical enters the body, is activated, metabolized, transported, and interacts with target sites [61]. Pesticides may be more lethal to *T. urticae* than to phytoseiids because the chemical is more readily activated in the pest and its target sites are highly sensitive. In contrast, predatory mites can often neutralize the chemical, or their target sites are less affected, rendering them more tolerant [61]. The mechanisms of toxicity and compatibility were not studied in these experiments. Findings from this study support the notion that even biorational pesticides can negatively affect certain natural enemies, although they may not always be directly toxic to specific species [31]. Mineral oil and citric acid caused the highest predator mortality (Figure 5). Importantly, the most efficacious product in this study, garlic oil, used against *T. urticae* and *A. cannabicola*, was not highly detrimental to *A. swirskii*. This contrasts with the findings of Busuulwa et al. [62], who reported a rapid decline in *A. swirskii* survival beyond 72 h of exposure to garlic oil. The authors, however, noted that garlic oil is primarily designed to repel insects from feeding on plants rather than to interfere with predator-prey interactions. This suggests that garlic oil has potential to be integrated into a management program together with *A. swirskii*. However, although mortality was low, oviposition should also be evaluated in future experiments to fully confirm their compatibility. Additionally, it is important to emphasize the effect of thyme oil, which proved highly efficacious in controlling *A. cannabicola* in the laboratory screening, while showing minimal adverse effects on *A. swirskii* survival. The proportion of live predatory mites remained consistently high after treatment application throughout the experimental period. These results are consistent with the findings of Abo-Taka et al. [63], who reported low mortality and high tolerance of *A. swirskii* to a 6% concentration of thyme oil, with an overall mortality of 66.53%. In conclusion, the botanical pesticides garlic oil and citric acid are approved for use on hemp in Florida and can be incorporated in IPM programs for the control of *T. urticae* infestations. In addition, these products offer extra advantages such as a minimal risk for beneficial organisms, being harmless to the environment, and human health. Our findings are consistent with those of Ismail et al. [64], indicating that garlic oil is a viable alternative to synthetic acaricides for controlling *T. urticae* populations in sustainable agricultural systems. Furthermore, *A. swirskii* is a promising biocontrol agent that has the potential to control both *T. urticae* and *A. cannabicola* even when garlic oil is applied. However, it would still be advisable to apply them at different times to minimize any direct negative effect of the oil on predator survival, considering that some oils may exert residual effects after application. Nonetheless, further studies are needed to evaluate the integration of biorational pesticides and predatory mites on hemp plants. Moreover, the evaluation and integration of chemical and biological control with other management strategies will lead to a more comprehensive and effective management program for the hemp industry in Florida.

Author Contributions: Conceptualization, M.A.C., L.M.S.A., L.S.O., S.L. and A.M.R.; methodology, M.A.C., L.M.S.A., P.V., M.D.G. and A.M.R.; software, M.A.C., L.M.S.A. and A.M.R.; validation, M.A.C. and L.M.S.A.; formal analysis, M.A.C., L.M.S.A. and A.M.R.; investigation, M.A.C., L.M.S.A., P.V. and M.D.G.; resources, L.S.O., N.T. and A.M.R.; data curation, M.A.C. and L.M.S.A.; writing—original

draft preparation, M.A.C.; writing—review and editing, all authors; visualization, M.A.C., L.M.S.A., S.L. and A.M.R.; supervision, S.L. and A.M.R.; project administration, N.T. and A.M.R.; funding acquisition, A.M.R. All authors have read and agreed to the published version of the manuscript.

Funding: This study was supported by the USDA-APHIS PPA7721 program (Fain # AP21PPQS&T00C064), the USDA-NIFA Research Capacity Fund (Hatch) project (Accession Number 7000779) and the University of Florida-USDA-ARS Non-Assistance Cooperative Agreement No. 58-6038-3-002. The mention of trade names or commercial products in this publication is solely for the purpose of providing specific information and does not imply recommendation or endorsement by the USDA. USDA is an equal opportunity provider and employer.

Data Availability Statement: The original data presented in the study are openly available in the figshare repository at <https://doi.org/10.6084/m9.figshare.30752603>.

Acknowledgments: The authors thank Zachary Brym and Ajit Williams (Agroecology Lab, Tropical Research and Education Center, Institute of Food and Agricultural Sciences, University of Florida) for their support in providing plants for the experiments, and Jose Alegria for his assistance during road trips to bring hemp plants from Apopka.

Conflicts of Interest: The authors declare that they have no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CBD	Cannabidiol
FDACS-DPI	Florida Department of Agriculture and Consumer Services-Division of Plant Industry
IPM	Integrated Pest Management
THC	Delta-9 tetrahydrocannabinol

References

- Small, E.; Marcus, D. Hemp: A new crop with new uses for North America. *Trends New Crops New Uses* **2002**, *24*, 284–326.
- Tancig, M.; Kelly-Begazo, C.; Kaur, N.; Sharma, L.; Brym, Z. Industrial hemp in the United States: Definition and history: SS-AGR-457/AG458, 9/2021. *EDIS* **2021**, *2021*, 1–3. [[CrossRef](#)]
- Williams, D.W.; Mundell, R. *An Introduction to Industrial Hemp and Hemp Agronomy*; University Kentucky Cooperative Extension Service: Lexington, KY, USA, 2018.
- Blare, T.; Rivera, M.; Ballen, F.H.; Brym, Z. Is a viable hemp industry in Florida's future? *EDIS* **2022**, *2022*, 1–5. [[CrossRef](#)]
- The National Agricultural Statistics Service (NASS); Agricultural Statistics Board; United States Department of Agriculture (USDA). National Hemp Report 2025. Available online: <https://share.google/eAPXInciZnf7MaCQW> (accessed on 29 August 2025).
- Fike, J. Industrial hemp: Renewed opportunities for an ancient crop. *Crit. Rev. Plant Sci.* **2016**, *35*, 406–424. [[CrossRef](#)]
- Ahmed, M.Z.; McKenzie, C.L.; Osborne, L.S. Arthropod and mollusk pests of hemp, *Cannabis sativa* (Rosales: Cannabaceae), and their indoor management plan in Florida. *J. Integr. Pest Manag.* **2024**, *15*, 1–22. [[CrossRef](#)]
- Buss, E.A.; Skelley, P.E. An initial list of arthropods on hemp (*Cannabis sativa* L.: Cannabaceae) in Florida. Florida Department of Agriculture and Consumer Services, Division of Plant Industry (FDACS-DPI, P002203). *Entomol. Circ.* **2023**, *445*, 1–12.
- McPartland, J.M. A survey of hemp diseases and pests. In *Advances in Hemp Research*; Haworth Press: New York, NY, USA, 1999; pp. 109–128.
- Pulkoski, M.; Burrack, H. Assessing the impact of piercing-sucking pests on greenhouse-grown industrial hemp (*Cannabis sativa* L.). *Environ. Entomol.* **2024**, *53*, 1–10. [[CrossRef](#)] [[PubMed](#)]
- Döker, İ.; Revynthi, A.M.; Mannion, C.; Carrillo, D. First report of acaricide resistance in *Tetranychus urticae* (Acari: Tetranychidae) from South Florida. *Syst. Appl. Acarol.* **2020**, *25*, 1209–1214. [[CrossRef](#)]
- Migeon, A.; Nougulier, E.; Dorkeld, F. Spider Mites Web: A Comprehensive Database for the Tetranychidae. Available online: <https://www1.montpellier.inrae.fr/CBGP/spmweb> (accessed on 29 August 2025).
- Gerson, U. Biology and control of the broad mite, *Polyphagotarsonemus latus* (Banks) (Acari: Tarsonemidae). *Exp. Appl. Acarol.* **1992**, *13*, 163–178. [[CrossRef](#)]

14. Skelley, P.E.; Buss, E.A.; Allen, J.S.; Bolton, S.J.; Deeter, L.A.; Fulton, J.C.; Greer, D.J.; Halbert, S.E.; Hayden, J.E.; Moore, M.R.; et al. Adventive arthropods in Florida as reported by the Florida Department of Agriculture and Consumer Services, Division of Plant Industry, 1990–2023. *Fla. Entomol.* **2025**, *108*, 20240027. [CrossRef]
15. Villanueva, R.; Vilorio, Z.; Ochoa, R.; Ulsamer, A. *Hemp Russet Mite, a Key Pest of Hemp in Kentucky*; University Kentucky Cooperative Extension Service: Lexington, KY, USA, 2022.
16. Szczepaniec, A.; Lathrop-Melting, A.; Janecek, T.; Nachappa, P.; Cranshaw, W.; Alnajjar, G.; Axtell, A. Suppression of hemp russet mite, *Aculops cannabicola* (Acari: Eriophyidae), in industrial hemp in greenhouse and field. *Environ. Entomol.* **2024**, *53*, 18–25. [CrossRef]
17. Bueno, A.F.; Panizzi, A.R.; Hunt, T.E.; Dourado, P.M.; Pitta, R.M.; Gonçalves, J. Challenges for adoption of integrated pest management (IPM): The soybean example. *Neotrop. Entomol.* **2021**, *50*, 5–20. [CrossRef] [PubMed]
18. Barzman, M.; Bärberi, P.; Birch, A.N.E.; Boonekamp, P.; Dachbrodt-Saaydeh, S.; Graf, B.; Hommel, B.; Jensen, J.E.; Kiss, J.; Kudsk, P.; et al. Eight principles of integrated pest management. *Agron. Sustain. Dev.* **2015**, *35*, 1199–1215. [CrossRef]
19. Fredricks, K.T.; Hubert, T.D.; Amberg, J.J.; Cupp, A.R.; Dawson, V.K. Chemical controls for an integrated pest management program. *N. Am. J. Fish. Manag.* **2021**, *41*, 289–300. [CrossRef]
20. Marčić, D.; Döker, İ.; Tsolakis, H. Bioacaricides in crop protection—What is the state of play? *Insects* **2025**, *16*, 95. [CrossRef]
21. Sharma, J.H.; Tayde, A.R. Evaluation of bio-rational pesticides, against brinjal fruit and shoot borer, *Leucinodes orbonalis* Guen. on brinjal at Allahabad agroclimatic region. *Int. J. Curr. Microbiol. Appl. Sci.* **2017**, *6*, 2049–2054. [CrossRef]
22. Florida Department of Agriculture and Consumer Services. Hemp and Pesticides Brochure. FDACS 2020. Available online: <https://cmedia.fdacs.gov/content/download/91491/file/hemp-pesticides-brochure.pdf> (accessed on 29 August 2025).
23. Nur, M.W.; Miah, M.R.U.; Amin, M.R. Management of aphid and pod borer of country bean using bio-rational pesticides. *Bangladesh J. Ecol.* **2020**, *2*, 1–5.
24. Nile, A.S.; Kwon, Y.D.; Nile, S.H. Horticultural oils: Possible alternatives to chemical pesticides and insecticides. *Environ. Sci. Pollut. Res.* **2019**, *26*, 21127–21139. [CrossRef] [PubMed]
25. Quesada, C.R.; Sadof, C.S. Efficacy of horticultural oil and insecticidal soap against selected armored and soft scales. *HortTechnology* **2017**, *27*, 618–624. [CrossRef]
26. Ismail, M.S.; Soliman, M.F.; Abo-Ghaila, A.H.; Ghallab, M.M. The acaricidal activity of some essential and fixed oils against the two-spotted spider mite in relation to different temperatures. *Int. J. Pest Manag.* **2015**, *61*, 121–125. [CrossRef]
27. Carrillo, D.; De Moraes, G.J.; Peña, J.E. (Eds.) *Prospects for Biological Control of Plant Feeding Mites and Other Harmful Organisms*; Springer International Publishing: Cham, Switzerland, 2015; Volume 19.
28. Knapp, M.; van Houten, Y.; van Baal, E.; Groot, T. Use of predatory mites in commercial biocontrol: Current status and future prospects. *Acarologia* **2018**, *58*, 72–82. [CrossRef]
29. Gerson, U.; Smiley, R.L.; Ochoa, R. *Mites (Acari) for Pest Control*; Blackwell Science: Oxford, UK, 2003; Volume 558, pp. 173–218.
30. Vechia, J.F.D.; Andrade, D.J.; Tassi, A.D.; Roda, A.; van Santen, E.; Carrillo, D. Can predatory mites aid in the management of the citrus leprosis mite? *Front. Agron.* **2023**, *5*, 1304656. [CrossRef]
31. Cloyd, R. Compatibility conflict: Is the use of biological control agents with pesticides a viable management strategy? Department of Natural Resources and Environmental Sciences, 384 National Soybean Research Laboratory, 1101 West Peabody Drive, Urbana, IL 61801, USA. 2005; *Unpublished Work*.
32. McMurtry, J.A.; De Moraes, G.J.; Sourassou, N.F. Revision of the lifestyles of phytoseiid mites (Acari: Phytoseiidae) and implications for biological control strategies. *Syst. Appl. Acarol.* **2013**, *18*, 297–320. [CrossRef]
33. Van Leeuwen, T.; Witters, J.; Nauen, R.; Duso, C.; Tirry, L. The control of eriophyoid mites: State of the art and future challenges. *Exp. Appl. Acarol.* **2010**, *51*, 205–224. [CrossRef] [PubMed]
34. Overmeer, W.P.J. Rearing and handling. In *Spider Mites: Their Biology, Natural Enemies and Control*; Elsevier: Amsterdam, The Netherlands, 1985; Volume 1, pp. 161–170.
35. Hartig, F.; Hartig, M.F. *Package ‘DHARMa’*; R Project: Vienna, Austria, 2017; Volume 531, p. 532.
36. Lenth, R.; Singmann, H.; Love, J.; Buerkner, P.; Herve, M. *Package “Emmeans”*. R Package Version 2018, 4.0–3. Available online: <http://cran.r-project.org/package=emmeans> (accessed on 29 August 2025).
37. Bates, D.; Mächler, M.; Bolker, B.; Walker, S. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* **2015**, *67*, 1–48. [CrossRef]
38. McPartland, J.M.; Clarke, R.C.; Watson, D.P. *Hemp Diseases and Pests: Management and Biological Control—An Advanced Treatise*; CABI Publishing: Wallingford, UK, 2000.
39. Azandeme-Hounmalon, G.Y.; Fellous, S.; Kreiter, S.; Fiaboe, K.K.M.; Subramanian, S.; Kungu, M.; Martin, T. Dispersal behavior of *Tetranychus evansi* and *T. urticae* on tomato at several spatial scales and densities: Implications for integrated pest management. *PLoS ONE* **2014**, *9*, e95071. [CrossRef]
40. Laska, A.; Rector, B.G.; Przychodzka, A.; Majer, A.; Zalewska, K.; Kuczynski, L.; Skoracka, A. Do mites eat and run? A systematic review of feeding and dispersal strategies. *Zool. J. Linn. Soc.* **2023**, *198*, 462–475. [CrossRef]
41. Bogan, C.E.; Ludwig, S.; Metz, B. *Using Oils as Pesticides*; Texas A&M AgriLife Extension: College Station, TX, USA, 2006.

42. Cranshaw, W.; Baxendale, B. *Insect Control: Horticultural Oils*; Colorado State University Extension Service: Fort Collins, CO, USA, 2005.
43. Mossa, A.T.H.; Afia, S.I.; Mohafrash, S.M.M.; Abou-Awad, B.A. Formulation and characterization of garlic (*Allium sativum* L.) essential oil nanoemulsion and its acaricidal activity on eriophyid olive mites (Acari: Eriophyidae). *Environ. Sci. Pollut. Res.* **2018**, *25*, 10526–10537. [\[CrossRef\]](#)
44. Roy, S.; Handique, G.; Barua, A.; Bora, F.R.; Rahman, A.; Muraleedharan, N. Comparative performances of jatropha oil and garlic oil with synthetic acaricides against red spider mite infesting tea. *Proc. Natl. Acad. Sci. India Sect. B Biol. Sci.* **2018**, *88*, 85–91. [\[CrossRef\]](#)
45. De Giosa, M.; Dale, A.G.; Wu, X.; Revynthi, A.M. Potential of commercial biorational and conventional pesticides to manage the Ruellia erinose mite in ornamental landscapes. *Insects* **2025**, *16*, 801. [\[CrossRef\]](#)
46. Attia, S.; Grissa, K.L.; Mailleux, A.C.; Lognay, G.; Heuskin, S.; Mayoufi, S.; Hance, T. Effective concentrations of garlic distillate (*Allium sativum*) for the control of *Tetranychus urticae* (Tetranychidae). *J. Appl. Entomol.* **2012**, *136*, 302–312. [\[CrossRef\]](#)
47. Prowse, G.M.; Galloway, T.S.; Foggo, A. Insecticidal activity of garlic juice in two dipteran pests. *Agric. For. Entomol.* **2006**, *8*, 1–6. [\[CrossRef\]](#)
48. Ribeiro, N.C.; da Câmara, C.A.G.; Melo, J.P.R.; de Moraes, M.M. Acaricidal properties of essential oils from agro-industrial waste products from citric fruit against *Tetranychus urticae*. *J. Appl. Entomol.* **2019**, *143*, 731–743. [\[CrossRef\]](#)
49. Helle, W.; Sabelis, M.W. (Eds.) *Spider Mites: Their Biology, Natural Enemies and Control*; Elsevier: Amsterdam, The Netherlands, 1985; Volume 1, pp. 141–160.
50. Lemay, J.; Scott-Dupree, C. Management of insect pests on cannabis in controlled environment production. In *Handbook of Cannabis Production in Controlled Environments*; CRC Press: Boca Raton, FL, USA, 2022; p. 275.
51. Duso, C.; Castagnoli, M.; Simoni, S.; Angeli, G. The impact of eriophyids on crops: Recent issues on *Aculus schlechtendali*, *Calepitrimerus vitis* and *Aculops lycopersici*. *Exp. Appl. Acarol.* **2010**, *51*, 151–168. [\[CrossRef\]](#)
52. Momen, F.M.; Abdel-Khalek, A. Effect of the tomato rust mite *Aculops lycopersici* (Acari: Eriophyidae) on the development and reproduction of three predatory phytoseiid mites. *Int. J. Trop. Insect Sci.* **2008**, *28*, 53–57. [\[CrossRef\]](#)
53. Vervaeke, L.; De Vis, R.; De Clercq, P.; Van Leeuwen, T. Is the emerging mite pest *Aculops lycopersici* controllable? Global and genome-based insights in its biology and management. *Pest Manag. Sci.* **2021**, *77*, 2635–2644. [\[CrossRef\]](#)
54. Park, H.H.; Shipp, L.; Buitenhuis, R. Predation, development, and oviposition by the predatory mite *Amblyseius swirskii* (Acari: Phytoseiidae) on tomato russet mite (Acari: Eriophyidae). *J. Econ. Entomol.* **2010**, *103*, 563–569. [\[CrossRef\]](#)
55. Mortazavi, N.; Fathipour, Y.; Talebi, A.A. The efficiency of *Amblyseius swirskii* in control of *Tetranychus urticae* and *Trialetrodes vaporariorum* is affected by various factors. *Bull. Entomol. Res.* **2019**, *109*, 365–375. [\[CrossRef\]](#) [\[PubMed\]](#)
56. Calvo, F.J.; Knapp, M.; van Houten, Y.M.; Hoogerbrugge, H.; Belda, J.E. *Amblyseius swirskii*: What made this predatory mite such a successful biocontrol agent? *Exp. Appl. Acarol.* **2015**, *65*, 419–433. [\[CrossRef\]](#) [\[PubMed\]](#)
57. Raman, A. The cannabis plant: Botany, cultivation and processing for use. In *Cannabis*; CRC Press: Boca Raton, FL, USA, 1998; pp. 35–40.
58. Van Houten, Y.M.; Glas, J.J.; Hoogerbrugge, H.; Rothe, J.; Bolckmans, K.J.F.; Simoni, S.; Van Arkel, J.; Alba, J.M.; Kant, M.R.; Sabelis, M.W. Herbivory-associated degradation of tomato trichomes and its impact on biological control of *Aculops lycopersici*. *Exp. Appl. Acarol.* **2013**, *60*, 127–138. [\[CrossRef\]](#) [\[PubMed\]](#)
59. Hayes, C. Developing Integrated Pest Management (IPM) Strategies for Hemp Russet Mite (*Aculops cannabicola* Farkas) on Hemp (*Cannabis sativa* L.). Master's Thesis, Colorado State University, Fort Collins, CO, USA, 2022.
60. Nomikou, M.; Janssen, A.; Schraag, R.; Sabelis, M.W. Phytoseiid predators as potential biological control agents for *Bemisia tabaci*. *Exp. Appl. Acarol.* **2001**, *25*, 271–291. [\[CrossRef\]](#) [\[PubMed\]](#)
61. Duso, C.; Van Leeuwen, T.; Pozzebon, A. Improving the compatibility of pesticides and predatory mites: Recent findings on physiological and ecological selectivity. *Curr. Opin. Insect Sci.* **2020**, *39*, 63–68. [\[CrossRef\]](#)
62. Busuulwa, A.; Riley, S.S.; Revynthi, A.M.; Liburd, O.E.; Lahiri, S. Residual effect of commonly used insecticides on key predatory mites released for biocontrol in strawberry. *J. Econ. Entomol.* **2024**, *117*, 2461–2474. [\[CrossRef\]](#)
63. Abo-Taka, S.M.; Sweelam, M.A.; Heikal, H.M.; Walash, I.H. Toxicity and biological activity of five plant extracts to the two-spotted spider mite, *Tetranychus urticae*, and predatory mite, *Amblyseius swirskii* (Tetranychidae: Phytoseiidae). *Acarines* **2014**, *8*, 49–56. [\[CrossRef\]](#)
64. Ismail, T.; Keratum, A.; El-Hetawy, L. Formulation of abamectin and plant oil-based nanoemulsions with efficacy against the two-spotted spider mite *Tetranychus urticae* (Acari: Tetranychidae) under laboratory and field conditions. *Appl. Biol. Chem.* **2022**, *65*, 61. [\[CrossRef\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.